UNIT 6 · STANDARD 6.NOS.4

# Prime Factorization

Lesson 6-13

**Name:** _____ **Date:** _____ **Class:** _____

**ORBITAL LOGISTICS MISSION**

## Cargo Codebreak

You are the logistics officer aboard Station Helios. A shipment of 60 supply crates just docked, and the sorting robots can only distribute cargo once it is broken into its prime building blocks. Master prime factorization and the station eats this week.

## My Notes

**Level 2 Standard**

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can write a number as a product of its prime factors using a factor tree.**

**Language Objective: I can explain how I broke a number down using the words prime number, composite number, factor, and exponent.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

**Prime Factorization** · **Prime number** · **Composite number** · **Factor** · **Factor tree** · **Exponent**

1. Writing a number as _____ numbers multiplied together is its prime factorization.

2 A whole number greater than 1 with exactly two factors, 1 and itself, is a _____ number.

3 A whole number greater than 1 that has more than two factors is a _____ number.

4 A number that divides evenly into another number is called a _____ .

5 I can break a number into its prime factors step by step using a _____ .

6 In $2^3$, the small number 3 is the _____ and it shows 2 is multiplied 3 times.

## Write About the Math

**Starting from its prime factorization, how can the gardener use the factors of 72 to decide which rectangular grids will use all 72 seed pods?**
*Empezando por su descomposición en factores primos, ¿cómo puede la jardinera usar los factores de 72 para decidir qué cuadrículas rectangulares usarán las 72 cápsulas de semillas?*

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Prime Factorization** *Factorización prima*
- ☐ **Prime number** *Número primo*
- ☐ **Composite number** *Número compuesto*
- ☐ **Factor** *Factor*
- ☐ **Factor tree** *Árbol de factores*

**Model:** *60 is composite because it has more than two factors.* — Students say 60 is composite because it has more than two factors (it is divisible by numbers other than 1 and 60, such as 2, 3, 5, 6), so it can be broken down.

## Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **Because 72 = ___, I can multiply combinations of those prime factors to get factors like ___, so a grid that works is ___ by ___.**
- **My answer is ___ because ___.**
  *Mi respuesta es ___ porque ___.*

## Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ___.**
  *Mi afirmación es ___.*
- **My evidence is ___, so ___.**
  *Mi evidencia es ___, entonces ___.*

## Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.